

```
CREATE TABLE hospital_data (  
  "Hospital Name" VARCHAR(100),  
  "Location" VARCHAR(100),  
  "Department" VARCHAR(100),  
  "Doctors Count" INT,  
  "Patients Count" INT,  
  "Admission Date" DATE,  
  "Discharge Date" DATE,  
  "Medical Expenses" DECIMAL(12,2)  
);  
SELECT * FROM hospital_data;
```

--Q1) Write a SQL query to find the total number of patients across all hospitals

```
SELECT SUM("Patients Count") AS total_patients  
FROM hospital_data;
```

--Q2) Retrieve the average count of doctors available in each hospital

```
SELECT "Hospital Name", ROUND(AVG("Doctors Count"), 2) AS average_doctors  
FROM hospital_data  
GROUP BY "Hospital Name"  
ORDER BY average_doctors DESC;
```

--Q3) Find top three hospital department that have the highest number of patients

```
SELECT "Department", SUM("Patients Count") AS total_patients  
FROM hospital_data  
GROUP BY "Department"  
ORDER BY total_patients DESC  
LIMIT 3;
```

--Q4)Identify the hospital that recorded the highest medical expenses

```
SELECT "Hospital Name", SUM("Medical Expenses") AS total_medical_expenses
FROM hospital_data
GROUP BY "Hospital Name"
ORDER BY total_medical_expenses DESC
LIMIT 1;
```

--Q5)Calculate the average medical expenses per day for each hospital

```
SELECT "Hospital Name",
ROUND(
SUM("Medical Expenses") /
SUM("Discharge Date" - "Admission Date"),
2
) AS average_daily_expenses
FROM hospital_data
GROUP BY "Hospital Name"
ORDER BY average_daily_expenses DESC;
```

```
ALTER TABLE hospital_data
ALTER COLUMN "Admission Date" TYPE DATE
USING TO_DATE("Admission Date", 'DD-MM-YYYY'),
ALTER COLUMN "Discharge Date" TYPE DATE
USING TO_DATE("Discharge Date", 'DD-MM-YYYY');
```

```
SELECT
"Admission Date",
"Discharge Date"
FROM hospital_data
LIMIT 5;
```

--Q6) Find the patient with the longest stay by calculating the difference between Discharged date and Admission date

```
SELECT "Hospital Name",  
"Location",  
"Department",  
"Admission Date",  
"Discharge Date",  
("Discharge Date" - "Admission Date") AS stay_days  
FROM hospital_data  
ORDER BY stay_days DESC  
LIMIT 1;
```

--Q7) Count the total number of patients treated in each city

```
SELECT "Location", SUM("Patients Count") AS total_patients  
FROM hospital_data  
GROUP BY "Location"  
ORDER BY total_patients DESC;
```

--Q8) Calculate the average number of days the patients spend in each department

```
SELECT "Department", ROUND(AVG("Discharge Date" - "Admission Date"), 2) AS average_stay_days  
FROM hospital_data  
GROUP BY "Department"  
ORDER BY average_stay_days DESC;
```

--Q9) Find the department with least numbers of patients

```
SELECT "Department", SUM("Patients Count") AS total_patients  
FROM hospital_data  
GROUP BY "Department"
```

```
ORDER BY total_patients ASC
```

```
LIMIT 1;
```

```
--Q10) Group the data by month and calculate the total expenses for each month
```

```
SELECT
```

```
TO_CHAR("Admission Date", 'YYYY-MM') AS month,
```

```
SUM("Medical Expenses") AS total_medical_expenses
```

```
FROM hospital_data
```

```
GROUP BY TO_CHAR("Admission Date", 'YYYY-MM')
```

```
ORDER BY month;
```